

9/22/26

Chapter 1: Sequence Alignment Algorithms

Learning Goal: Learn about the combinatorics of sequences (prefixes, suffixes, etc.) and how dynamic programming allows all of this to work

- 1.1 Edit Graphs, Dynamic Programming, Principle of Optimality
- 1.2 GLOBAL ALIGNMENT ALGORITHM
- 1.3 Heuristic interpretation of alignment score
- 1.4 Scoring schema for alignment → mathematical models for evolution
- 1.5 LOCAL ALIGNMENT ALGORITHM

1.5 Local Alignment Algorithms

Combinatorics of sequences/strings/words

Prefixes

Suffixes

Substrings

$X = \text{AACTT}$

Substrings(X)

$\text{Pref}(X) = A$

AA

AAC

AACT

AACTT

$\text{Suff}(X) = T$

TT

CTT

ACTT

AACTT

Position Intervals

Substrings(X) =

A	1-1
AA	1-2
AAC	1-3
AACT	1-4
AACTT	1-5
A	2-2
AC	2-3
ACT	2-4
ACTT	2-5
C	3-3

CT	3-4
CTT	3-5
T	4-4
TT	4-5
T	5-5

$\text{Suffixes}(\text{Prefixes}(X)) = \text{Substrings}(X)$

$\text{Prefixes}(\text{Suffixes}(X)) = \text{Substrings}(S)$

Pastiche Pie is given to the top 10 students in this class, and they get hired as TAs

The Local Alignment Problem

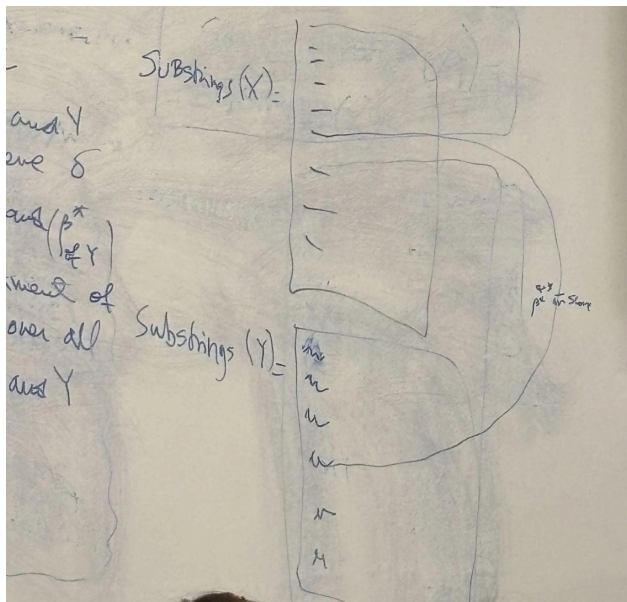
Given: Two sequences X and Y and a scoring scheme δ

Compute: two substrings (α^* of X) and (β^* of Y) whose optimal global alignment of α^* and β^* is maximized over all pairs of substrings of X and Y

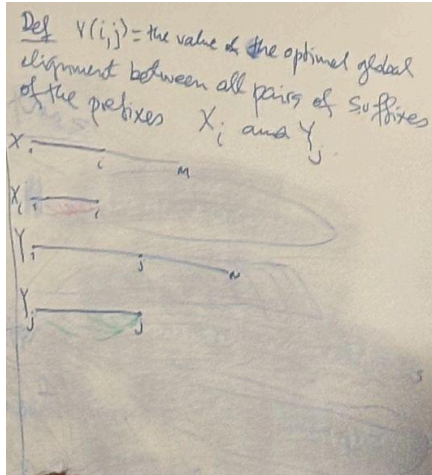
That alignment $\alpha^* \rightarrow \beta^*$

$\beta^* \rightarrow \alpha^*$

Is the optimal local alignment



Def: $v(i,j)$ = value of optimal global alignment between all pairs of suffixes of the prefixes x_i and y_j



The following algorithm computes $V^{\text{opt-local-align}}(X,Y) = V^{\text{OLA}}(X,Y)$

$V(i, 0) = 0, V(0,1) = 0$

FOR $i > 0$ AND $j > 0$ DO

$V(i,j) = \text{MAX} \{$
 0
 $V(i-1,j-1) + \delta(x_i, y_j),$
 $V(i-1,j) + \delta(x_i, -),$
 $V(i,j-1) + \delta(-, y_j)\}$

$V^{\text{OLA}}(X,Y) = \text{MAX}[V(i,j), 0 \leq i \leq N \quad 0 \leq j \leq M]$



How come the algorithm stops at the substring

Proof of algorithm

Let α be a suffix of X , and β be a suffix of Y . α^* and β^* as the optimal local alignment of X and Y

- 1) Since α or β can be empty ϵ then 0 is a value of the alignment

- 2) Suppose α^* and β^* are not empty strings then letter $X(i) = x_i$ must be aligned with “-” or with $Y(j) = y_j$ similarly for Y

The empty string E = string with infinite symbols and letters

